SIMATS ENGINEERING

Lab Programs

COURSE CODE: ECA1009 **COURSE NAME: Microprocessors & Microcontrollers**

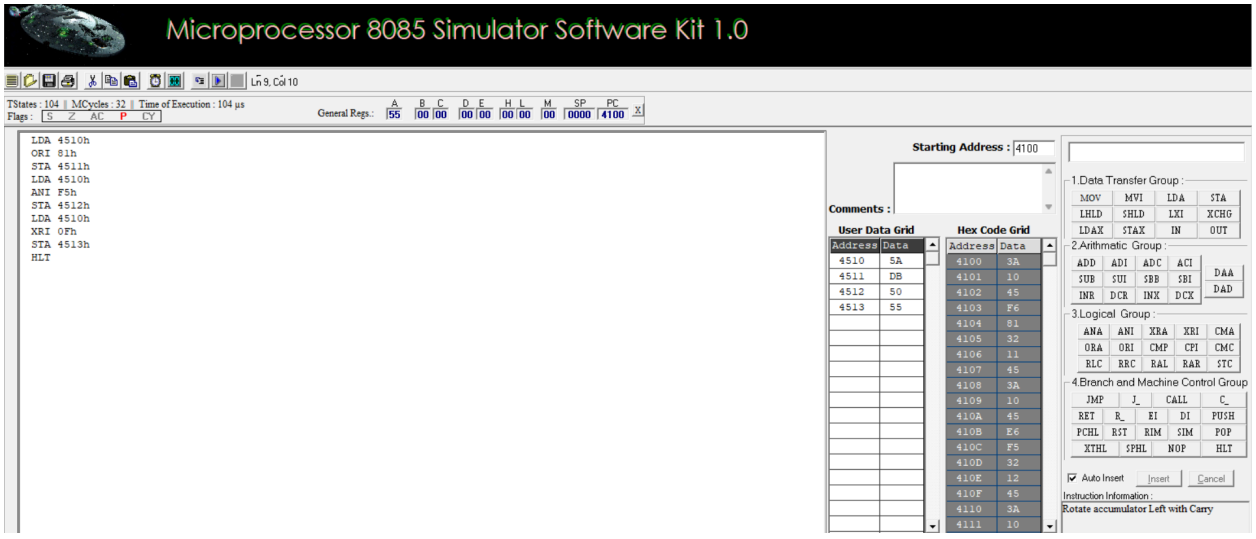
Program B – Testing of Bit Setting and Clearing Operations Using 8-Bit Assembly Language Programming

Aim: To implement and test bit setting and clearing operations using 8-bit Intel 8085 assembly language programming.

Algorithm:

1. Load the 8-bit data from memory location 4510H into the accumulator.
2. Perform the bit setting operation using the `ORI 81H` instruction.
3. Store the result at memory location 4511H.
4. Load the original data and perform the bit clearing operation using the `ANI F5H` instruction.
5. Store the result at memory location 4512H.
6. Load the original data, perform the bit toggling operation using `XRI 0FH`, store the result at 4513H, and halt.

Execute:



Result: The bit setting, clearing, and toggling operations were executed successfully using the 8085 simulator, and the results were stored in the specified memory locations.